

Lecture 2

$$\frac{1}{L} \sum_{i=1}^L [a(x_i) \neq y_i] \rightarrow \min_a$$

$y_i \in \{1, \dots, K\}$ - classes

$$P = \begin{pmatrix} P(\hat{y}_i = 1) \\ P(\hat{y}_i = 2) \\ \vdots \\ P(\hat{y}_i = K) \end{pmatrix}$$

$$z_j = \sigma(W_j z_{j-1} + b_j)$$

$$\vdots$$
$$z_n = W_n z_{n-1} + b_n$$

$$z_n \in \mathbb{R}^{d_n}$$

$$P = \{p_k\}_{k=1}^K$$

$$1. p_k \geq 0$$

$$2. \sum p_k = 1$$

$$P(\hat{y} | x)$$

$$H(y|x) = - \sum_{k=1}^K p[y=k|x] \ln p[y=k|x]$$

$$p_k \rightarrow p[y=k|x] \begin{bmatrix} 0, 0, \dots, 1, 0, 0 \end{bmatrix}^T$$

y_{true}

$$1. \sum_{k=1}^K p_k - 1 = 0$$

$$\left| \begin{array}{l} f(x, k) \in \mathbb{R}^d \\ f(x, [1..K]) \in \mathbb{R}^{K \times d} \end{array} \right.$$

$$2. \sum_{k=1}^K p_k f(x, k) =$$

$$= \mathbb{E}[f(x, k) | x] \quad \mu \in \mathbb{R}^d$$

$$\mathcal{L}(p, \lambda, \mu) = - \sum_{k=1}^K p_k \ln p_k +$$

$$+ \lambda \left(\sum_{k=1}^K p_k - 1 \right) + \mu^T \left(\sum_{k=1}^K p_k f(x, k) -$$

$$- \mathbb{E}[f(x, k) | x] \right)$$

$$\frac{\partial \mathcal{L}}{\partial p_k} = -\ln p_k - 1 + \lambda + \mu^T f(x, k) = 0$$

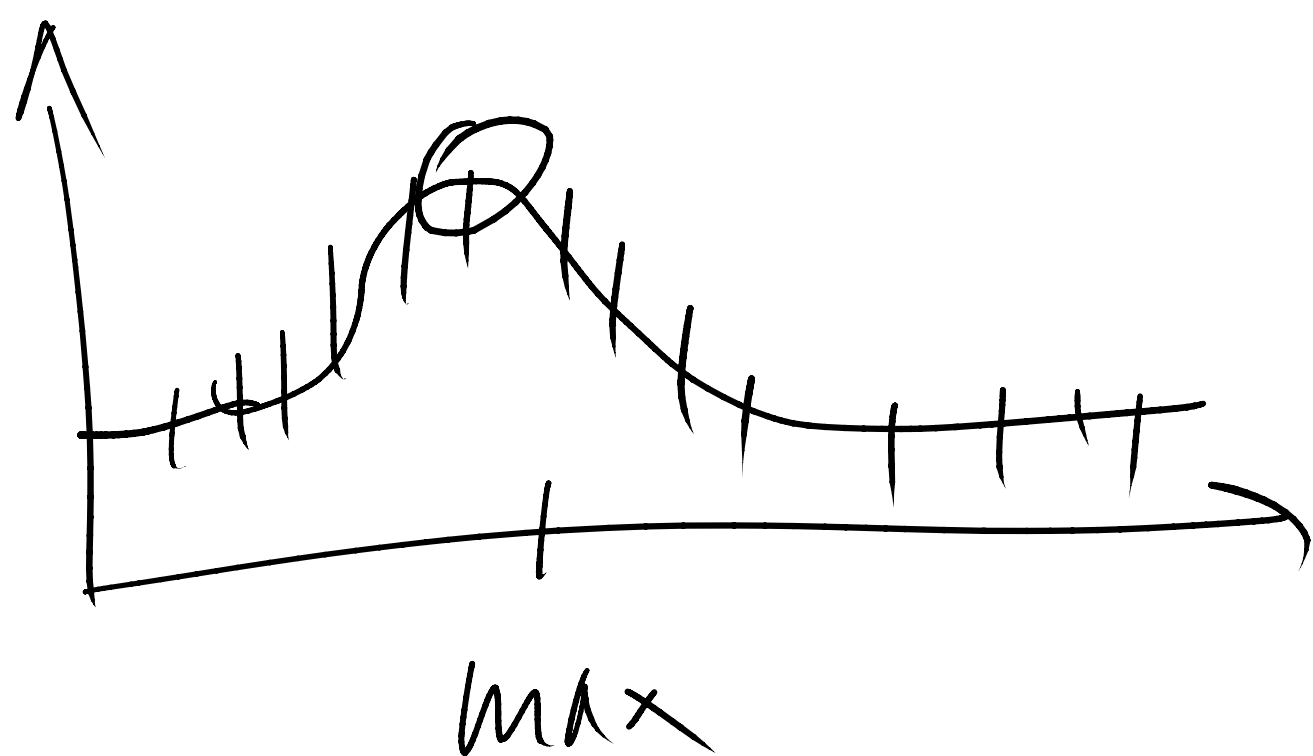
$$p_k = \frac{\exp(\mu^T f(x, k))}{\exp(1 - \lambda)} = z(x)$$

$$\sum_{j=1}^K p_k = \frac{\sum_{j=1}^K \exp(\mu^T f(x, j))}{Z(x)} = 1$$

$$Z(x) = \sum_{j=1}^K \exp(\underbrace{\mu^T f(x, j)}_{z_j})$$

$$p_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}$$

- softmax
argmax



z_k - logits

p_k - probas

$i, j \quad z_i > z_j \quad p_i > p_j \quad \text{monotone}$

MLE

y - target $\in \{1, \dots, K\}$

$$-\sum_{k=1}^K [k=y] \log p_k \rightarrow \min_{\{W, b\}} \\ \underbrace{\hspace{10em}}_{-\log p_y}$$

Negative Likelihood Loss

$$\{(x_i, y_i)\}_{i=1}^l$$

$$-\sum_{i=1}^l \sum_{k=1}^K [k=y_i] \log p_k^{(i)} \rightarrow \min_{W, b}$$

$$-\sum_{i=1}^l \log p_{y_i}^{(i)} \rightarrow \min \quad \sim \quad \prod_{i=1}^l p_{y_i}^{(i)} \rightarrow \max$$

NLL loss (softmax(z), y) → min

log exp

Cross Entropy Loss

$$\log \text{softmax}(z_i) = \log \frac{e^{z_i}}{\sum_j e^{z_j}} =$$

$$= \log e^{z_i} - \log \sum_j e^{z_j} = z_i -$$

$$- \log \left(\sum_j e^{z_j} \right)$$

$$\log \text{softmax} = \log \frac{\exp(z_i - \max_k z_k)}{\sum_j \exp(z_j - \max_k z_k)}$$

VAT (Universal Approximation Theorem)

Cybenko, 1989

f -continuous

σ -Sigmoid

$$\left\{ \begin{array}{l} \lim_{x \rightarrow -\infty} \sigma(x) = 0 \\ \lim_{x \rightarrow +\infty} \sigma(x) = 1 \end{array} \right.$$

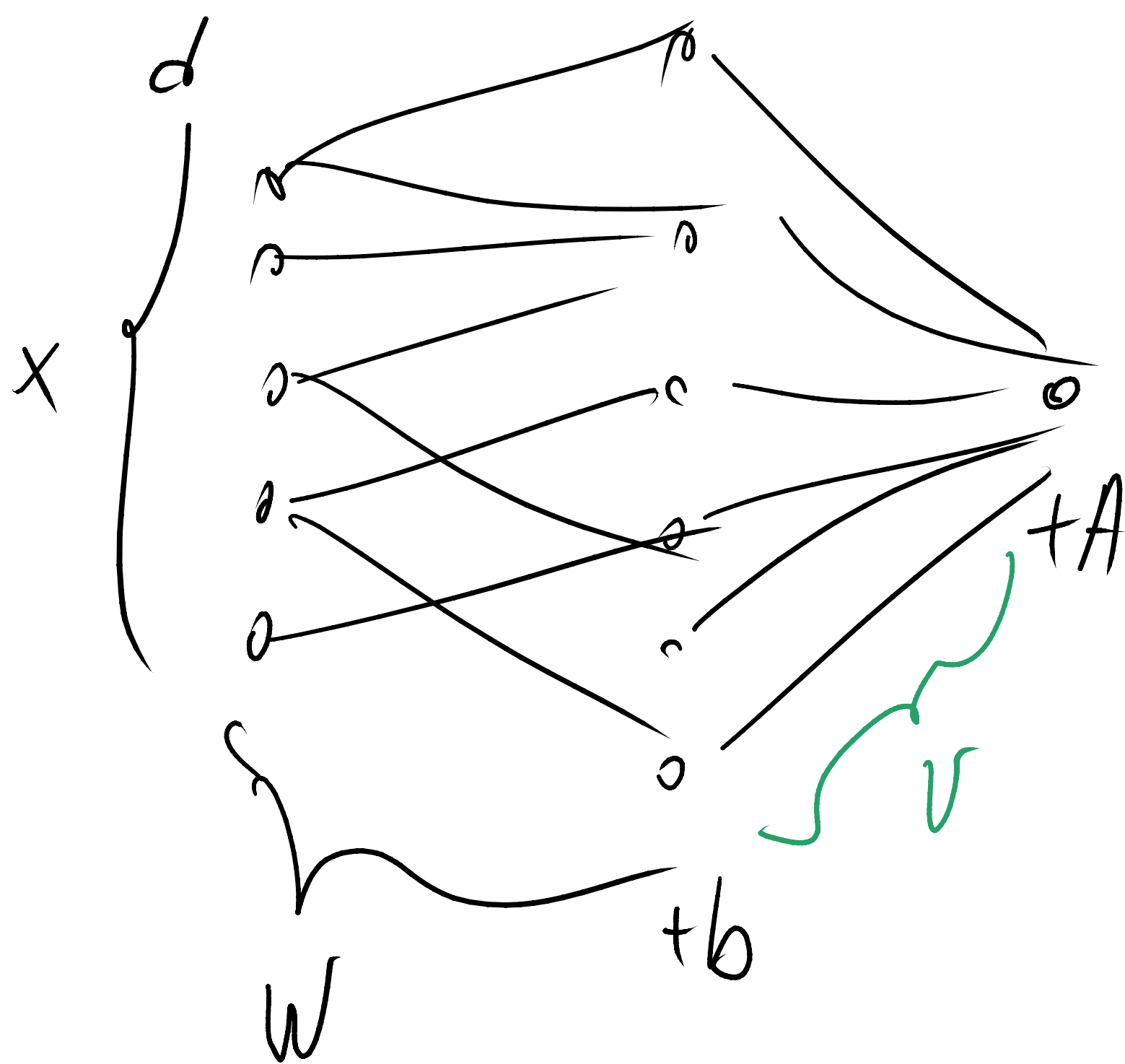
$$\lim_{x \rightarrow +\infty} \sigma(x) = 1$$

$$\forall \varepsilon > 0, \forall f \in C([0, 1]^d) \quad f: \mathbb{R}^d \rightarrow \mathbb{R}$$

$$\exists n \in \mathbb{N}, w \in \mathbb{R}^{d \times n} \quad w_i \in \mathbb{R}^d \quad b \in \mathbb{R}^d$$

$$V \in \mathbb{R}^n \quad v_i \in \mathbb{R}, A \in \mathbb{R}$$

$$g(x) = \sum_{i=1}^n v_i \sigma(w_i^T x + b_i) + A$$



$$\sup_{x \in [0,1]^d} |f(x) - g(x)| \leq \epsilon$$

Dropout $z_k \in \mathbb{R}^{B \times d_k}$

B - batch size

d_k - hidden size

$$m \in \{0, 1\}^{B \times d_k}$$

$$m_{ij} \sim \text{Bern}(1-p)$$

$$z_{k+1} = m \odot z_k \frac{1}{1-p} \quad \text{element-wise}$$

$$z_{k+1}^{\text{Inf}} = z_k$$

eval mode

Batch Normalization

$$(-\infty, +\infty)$$

0 mean 1 std

$$z_k \in \mathbb{R}^{B \times d_k}$$

$$z_k^i$$

$$\left(\overset{\downarrow z_k^i}{-} \right) - z_k$$

$$\mu = \frac{1}{B} \sum_{i=1}^B z_k^i$$

$$\sigma^2 = \frac{1}{B} \sum_{i=1}^B (z_k^i - \mu)^2$$

$$\mu, \sigma^2 \in \mathbb{R}^{d_k}$$

$$\alpha_k = \frac{z_k - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

$$z_{k+1} = \alpha_k \odot w + b$$

$$w, b \in \mathbb{R}^{d_k}$$

$$\text{running_mean} = \text{running_mean} (1-m) + m \cdot \mu \frac{\beta}{\beta-1}$$

$$\text{running_var} = \text{running_var} (1-m) + \sigma^2 m$$

$$m \approx 0.1$$

$$\alpha_k = \frac{z_k - \text{running_mean}}{\sqrt{\text{running_var} + \epsilon}} \quad \text{eval}$$

Classification

1. Label smoothing

$$(0, 0, 0, \dots, \underset{\substack{\uparrow \\ \text{y_true}}}{1}, \dots, 0, 0) = y$$

k -classes

$$\hat{y} = \frac{y + \alpha}{1 + K\alpha}$$

$$(\alpha, \dots, \alpha, 1-\beta, \alpha, \dots)$$

2. Mixup

$$z_1, z_2 - \text{logits for } x_1, x_2$$

y_1, y_2

$$\alpha X_1 + (1-\alpha) X_2$$

$$\alpha y_1 + (1-\alpha) y_2$$